Code-Layout, Readability and Reusability

Date	01 JULY 2026
Team ID	XXXXXX
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing and indentation are maintained throughout the code.	Yes	Follows Python (PEP 8) coding standards.
2	Proper File Structure	Backend, frontend, models, routes, and database files are logically organized.	Yes	Project uses a modular folder structure.
3	Meaningful Variable Names	Variables clearly represent their purpose.	Yes	Descriptive names improve readability.
4	Function / Method Names	Functions are named according to their functionality.	Yes	Easy to understand and maintain.
5	Code Comments	Important logic is documented with comments where necessary.	Partial	Additional comments can improve maintainability.
6	Modular Design	Code is divided into reusable modules and services.	Yes	Authentication, loan management, and prediction modules are separated.
7	No Redundant Code	Duplicate or unused code is minimized.	Yes	Reusable functions reduce redundancy.
8	Error Handling	Exceptions and validation errors are handled properly.	Yes	FastAPI exception handling and input validation are implemented.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technolog	yWhere Reused	Reusability Level (High / Medium / Low)
1	Authentication Module	Python, FastAPI	User Login & Registration	High
2	Loan Management Module	Python, SQLAlchemy	Loan CRUD Operations	High
3	Financial Health Analysis Module	Python	Financial Score Calculation	High
4	Debt Settlement Prediction Module	Python, AI Logic	Settlement Recommendation	High
5	Database Models	SQLite, SQLAlchemy	Shared Across Backend Services	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized backend and frontend with clear separation of responsibilities.
Readability	5	Meaningful naming conventions and clean coding practices improve readability.
Reusability	5	Modular architecture allows components to be reused across different features.
Documentation / Comments	4	Basic documentation is available; adding more inline comments would further improve maintainability.
Overall Score	4.8 / 5	The project follows good software engineering practices with a modular, maintainable, and scalable codebase.